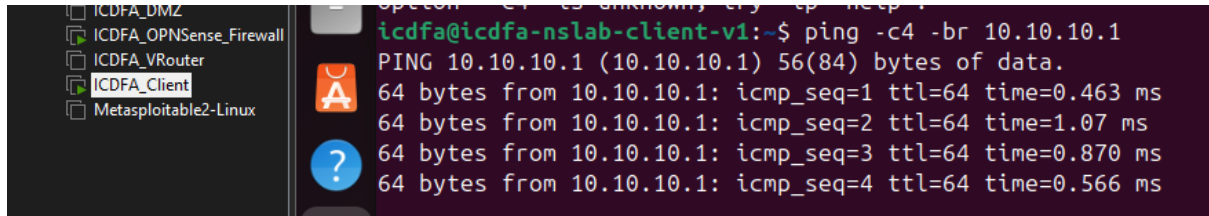


FULL NAME	MUSA SHUAIBU
REGISTRATION NUMBER	C11/26/FCDF/17153
UBUNTU	ICDFA provided
VM HOSTNAME	Icdfa-nslab-client-v1

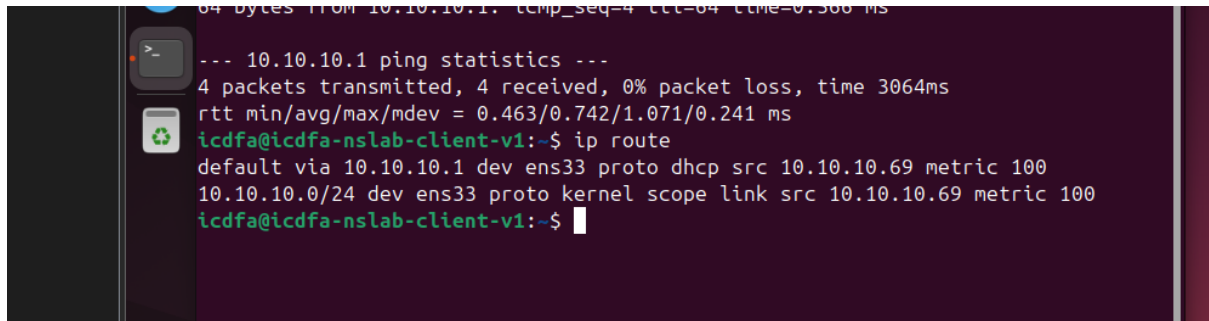
PART A: RECORDING THE BASELINE

1.0 Testing the connectivity



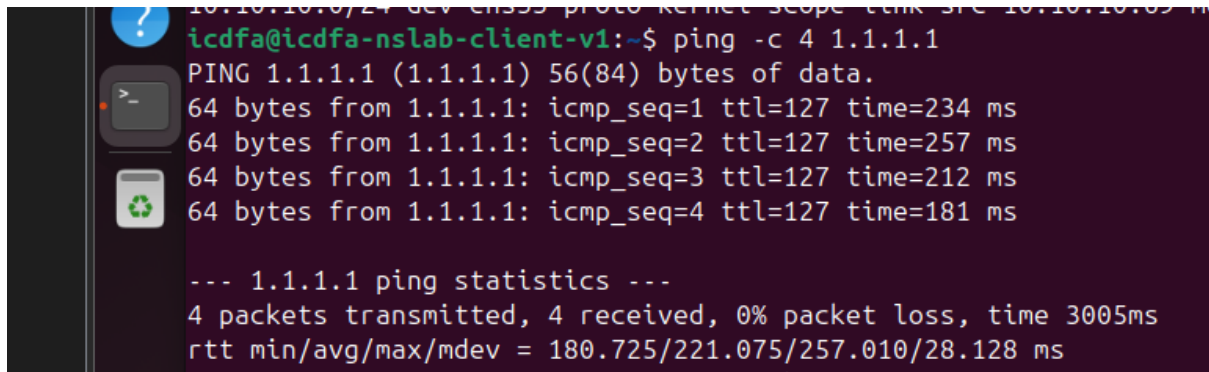
```
icdfa@icdfa-nslab-client-v1:~$ ping -c4 -br 10.10.10.1
PING 10.10.10.1 (10.10.10.1) 56(84) bytes of data.
64 bytes from 10.10.10.1: icmp_seq=1 ttl=64 time=0.463 ms
64 bytes from 10.10.10.1: icmp_seq=2 ttl=64 time=1.07 ms
64 bytes from 10.10.10.1: icmp_seq=3 ttl=64 time=0.870 ms
64 bytes from 10.10.10.1: icmp_seq=4 ttl=64 time=0.566 ms
```

Brief Explanation: This shows that, the client VM can communicate with the default gateway of the firewall 10.10.10.1



```
icdfa@icdfa-nslab-client-v1:~$ ip route
default via 10.10.10.1 dev ens33 proto dhcp src 10.10.10.69 metric 100
10.10.10.0/24 dev ens33 proto kernel scope link src 10.10.10.69 metric 100
icdfa@icdfa-nslab-client-v1:~$
```

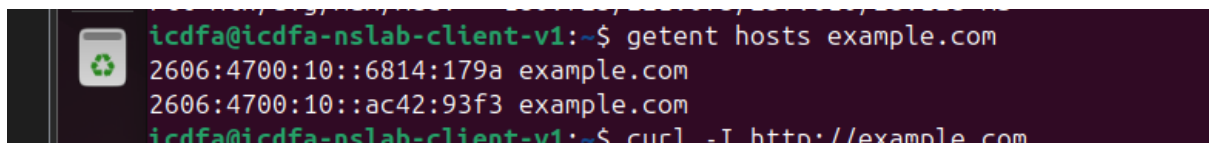
Brief Explanation: This shows the default gateway 10.10.10.1 is learn by dhcp sever the client ip address assign by the dhcp server is 10.10.10.69



```
icdfa@icdfa-nslab-client-v1:~$ ping -c 4 1.1.1.1
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data.
64 bytes from 1.1.1.1: icmp_seq=1 ttl=127 time=234 ms
64 bytes from 1.1.1.1: icmp_seq=2 ttl=127 time=257 ms
64 bytes from 1.1.1.1: icmp_seq=3 ttl=127 time=212 ms
64 bytes from 1.1.1.1: icmp_seq=4 ttl=127 time=181 ms

--- 1.1.1.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 180.725/221.075/257.010/28.128 ms
```

Brief Explanation: This show that the client VM can be able to access the internet through the WAN interface of the OPNSense firewall



```
icdfa@icdfa-nslab-client-v1:~$ getent hosts example.com
2606:4700:10::6814:179a example.com
2606:4700:10::ac42:93f3 example.com
icdfa@icdfa-nslab-client-v1:~$ curl -I http://example.com
```

Brief Explanation: This show that the DNS server on the firewall can resolve a domain name to an ip address

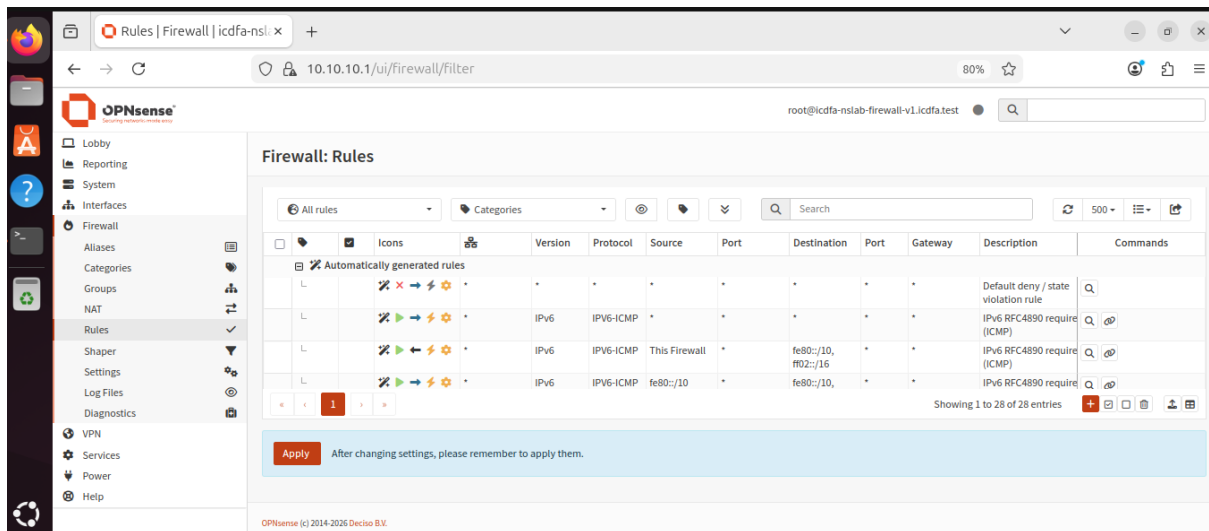
```
2006:4700:10::ac42:9313 example.com
icdfa@icdfa-ns-lab-client-v1:~$ curl -I http://example.com
HTTP/1.1 200 OK
Date: Tue, 06 Oct 2026 07:07:30 GMT
Content-Type: text/html; charset=utf-8
Connection: keep-alive
Server: cloudflare
Last-Modified: Sun, 04 Oct 2026 20:44:03 GMT
Allow: GET, HEAD
Accept-Ranges: bytes
Age: 10538
cf-cache-status: HIT
CF-RAY: a462d458e82e9f94-AMS
alt-svc: h3=":443"; ma=86400
```

Brief Explanation: This shows the client can received respond from the http request

```
icdfa@icdfa-ns-lab-client-v1:~$ curl -I https://example.com
HTTP/2 200
date: Tue, 06 Oct 2026 07:07:46 GMT
content-type: text/html; charset=utf-8
server: cloudflare
last-modified: Fri, 02 Oct 2026 16:11:02 GMT
allow: GET, HEAD
accept-ranges: bytes
age: 10552
cf-cache-status: HIT
cf-ray: a462d4bdace70b64-AMS
alt-svc: h3=":443"; ma=86400
```

Brief Explanation: This shows the client can received respond from the https request which is more secure and packet in transit are encrypted using TLS/SSL

PARR B: REVIEW RULE ORDER



Brief Explanation: This part explains why block rule have to come above the top rule assuming we have a rule like

Top

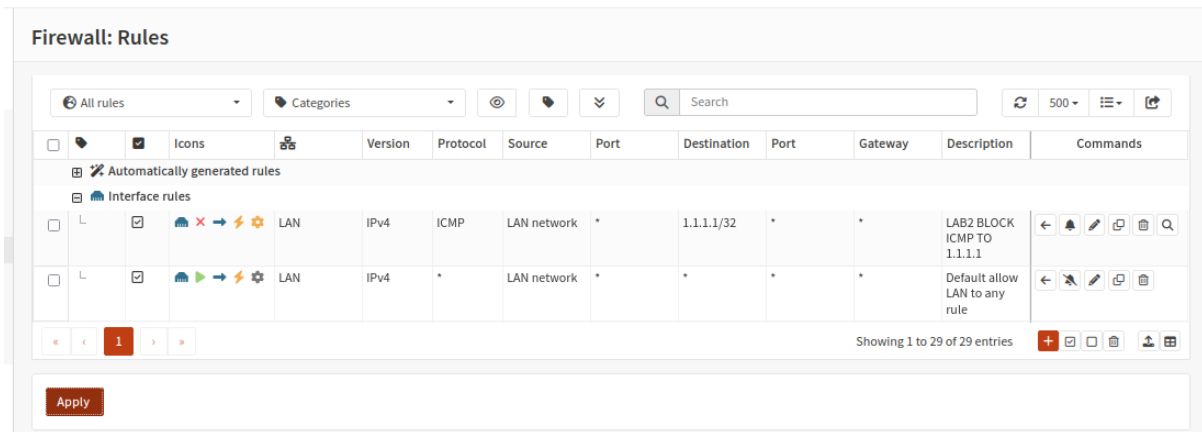
Allow pass LAN any

Bottom

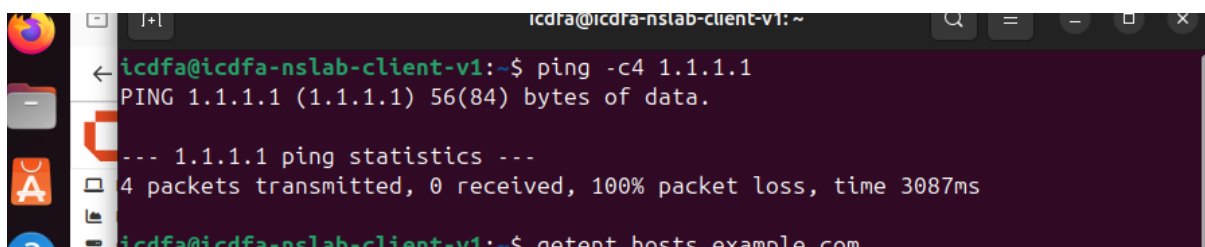
Block ICMP to 10.10.10.1

Since the top has already allow the block cannot filter the packet that is the reason why block comes at the top before allow

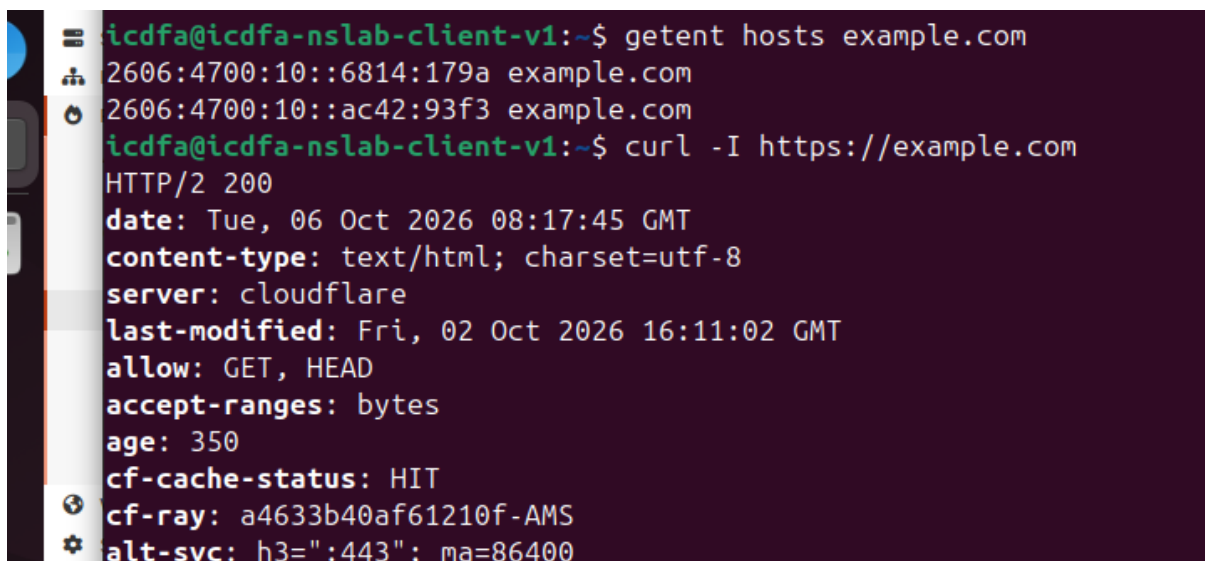
PART C: BLOCKING ICMP



Brief Explanation: This new block rule added is below the permit rule of the LAN interface therefore there is need to move it above the permit rule which can be done using the check box and the arrow key

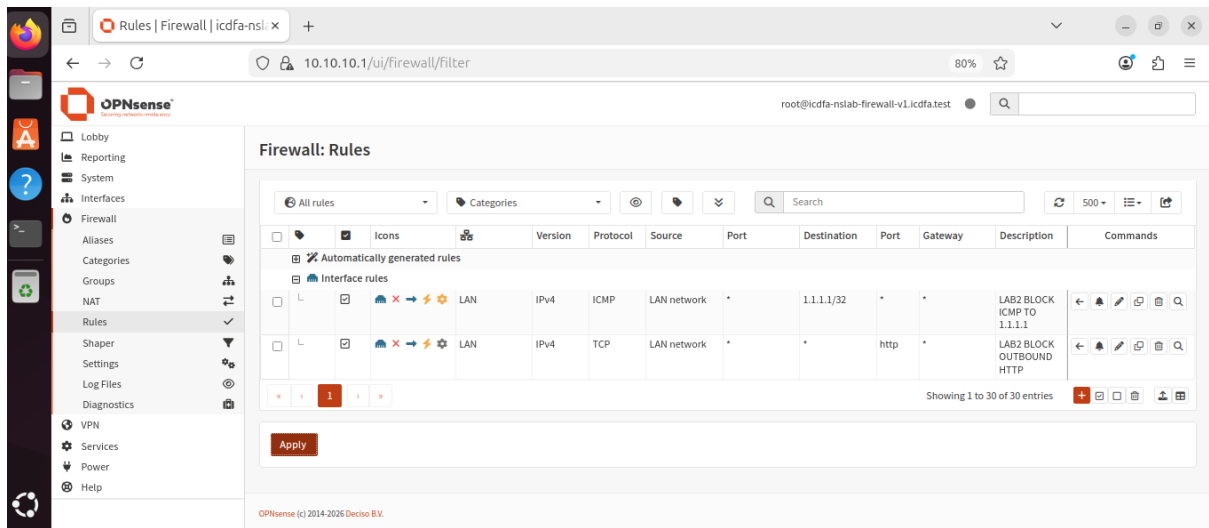


Brief Explanation: This shows that the new rule define has successfully block the ICMP request to destination 1.1.1.1

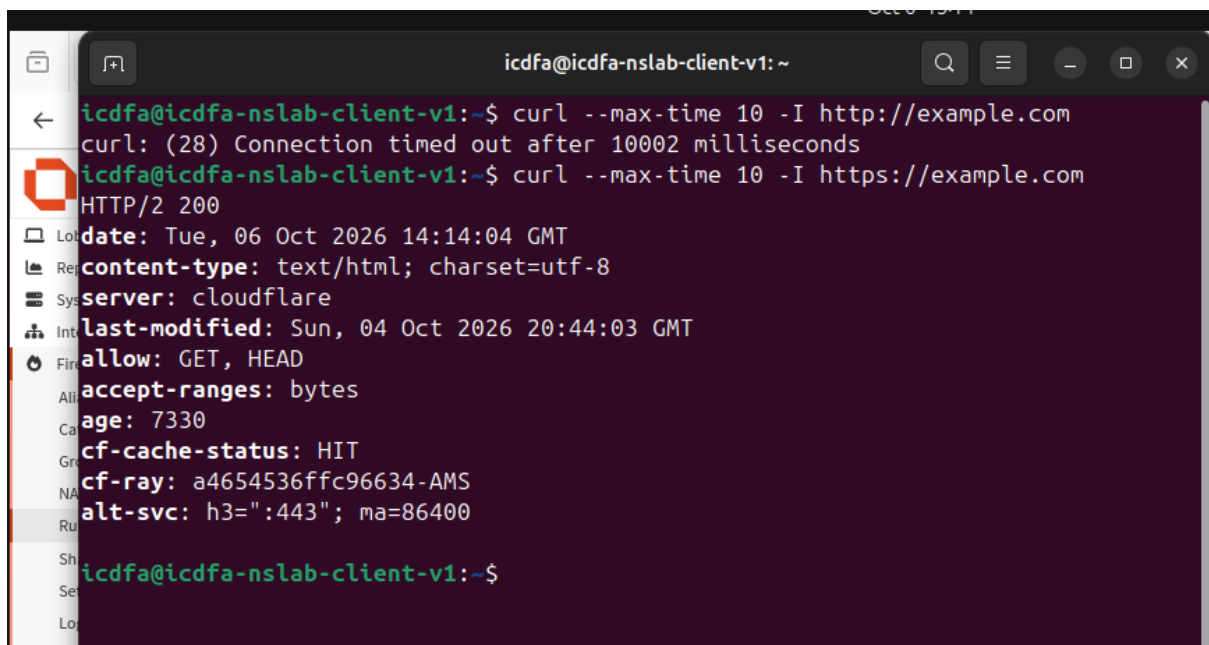


Brief Explanation: This shows that the new rule define does not affect the DNS protocol because the rule is defined specifically to ICMP protocol not DNS, HTTP or HTTPS protocol

PART D: BLOCKING OUTBOUND HTTP WHILE ALLOWING HTTPS

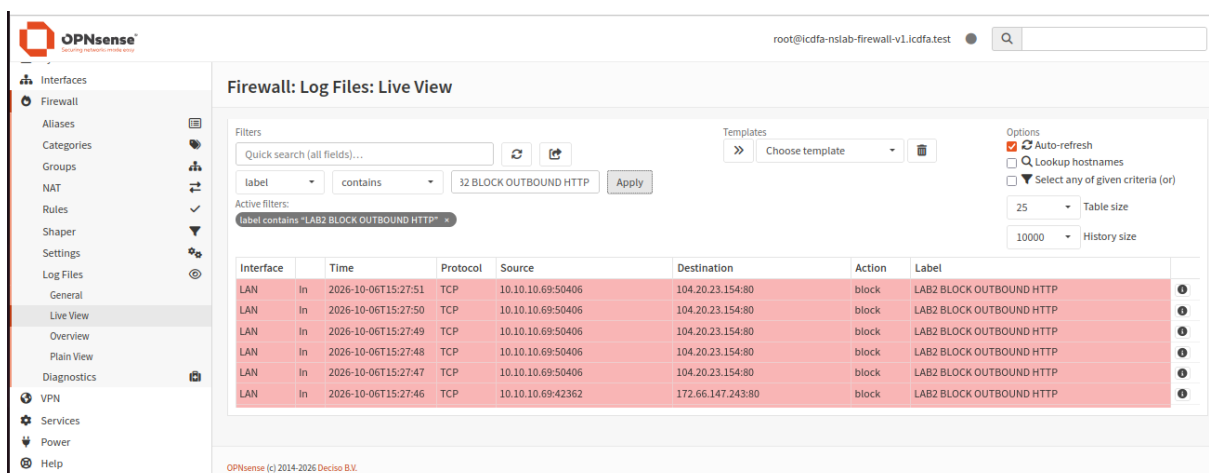


Brief Explanation: This shows the rule created to block the TCP protocol of any user that wanted to access HTTP which is not secured website



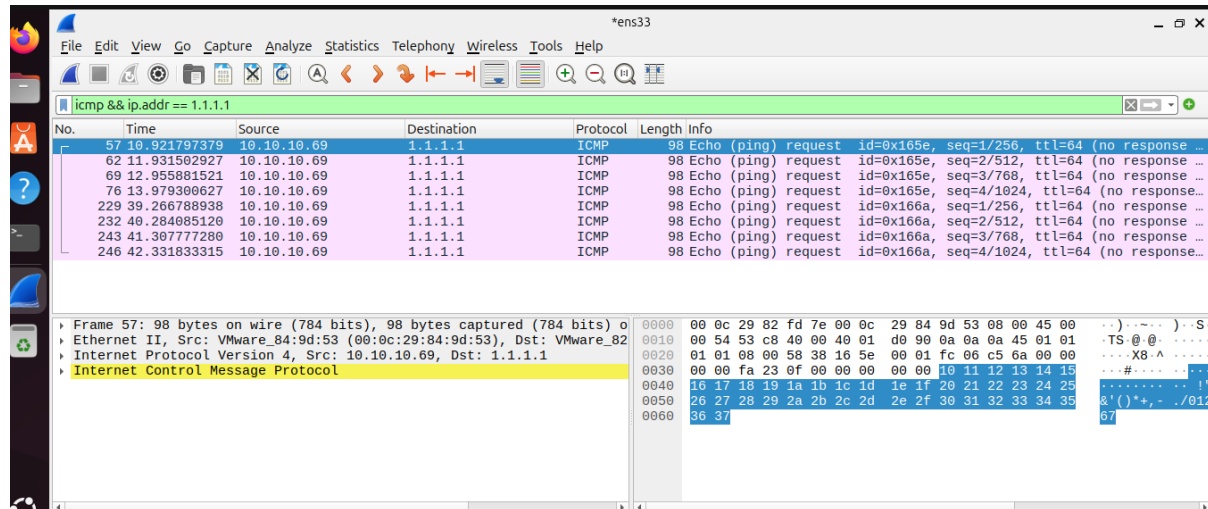
Brief Explanation: This shows that ubuntu client was denied from accessing the HTTP website but can access the HTTPS website the block rule applied has work successfully

PART E: EXAMIN FIREWALL LOGS

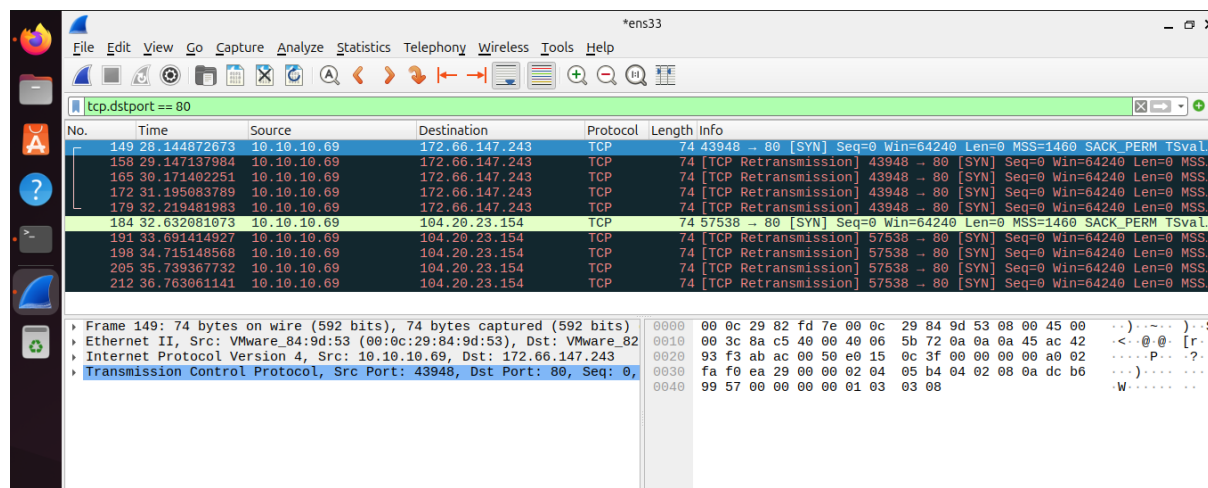


Brief Explanation: the traffic matched the block rule instead of the default allow rule because the block rule is been move to the top. The firewall will first check it hierarchy from block rule first before the permit rule which is top to bottom

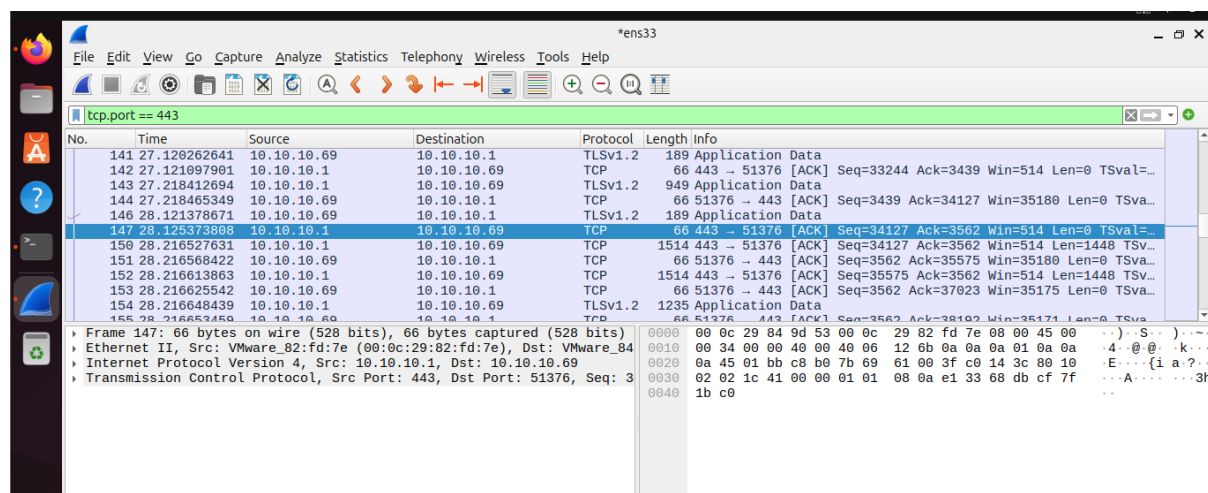
PART F: CORELATING WITH WIRESHARK



Brief Explanation: by using the icmp and ip addr filter the Wireshark was able to captured the request that is bee sent to the destination 1.1.1.1 but no any reply was responded because of the block policy that is been define for ICMP



Brief Explanation: by using the tcp.dstport filter the Wireshark was able to captured the TCP SYN request that is bee sent but no any ACK reply was responded because of the block policy that is been define for tcp protocol while trying to reach HTTP



Brief Explanation: by using the `tcp.dstport == 443` filter the Wireshark was able to capture the TCP SYN request that is being sent and, in this case, it responds with ACK reply because there is no any block policy that is been define for tcp protocol while trying to reach HTTPS

PART G: INSPECT STATES AND AUTOMATIC NAT

root@icdfa-ns-lab-firewall-v1.icdfa.test

Firewall: Diagnostics: States

Select rule

States Actions

Search

☐	Int	Dir	Proto	Source	Nat	Destination	Gateway	State	Rule	Commands
☐	all	→	tcp	10.10.10.69:51376		10.10.10.1:443		TIME_WAIT:TIME_W...	anti-lockout rule	
☐	all	→	tcp	10.10.10.69:56380		34.107.243.93:443		ESTABLISHED:ESTA...	Default allow LAN to...	
☐	all	←	tcp	192.168.163.132:9063	10.10.10.69:56380	34.107.243.93:443		ESTABLISHED:ESTA...	let out anything fro...	
☐	all	→	udp	10.10.10.69:57121		10.10.10.1:53		SINGLE:MULTIPLE	Default allow LAN to...	
☐	all	→	udp	10.10.10.69:37878		10.10.10.1:53		SINGLE:MULTIPLE	Default allow LAN to...	
☐	all	←	udp	192.168.163.132:12598		162.159.44.228:53		MULTIPLE:SINGLE	let out anything fro...	

Showing 1 to 25 of 25 entries

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Brief Explanation: For the LAN side we can see Ubuntu private IP = 10.10.10.69, temporary source port selected by Ubuntu = 56380, remote HTTPS server = 34.107.243.93 and HTTPS destination port = 443 from the WAN side we can see that **Nat (192.168.163.132:9063 10.10.10.69:56380 34.107.243.93:443)** is the address and port to which the session is translated using NAT

Firewall: NAT: Source NAT

Mode Automatic Source NAT rule generation

Categories

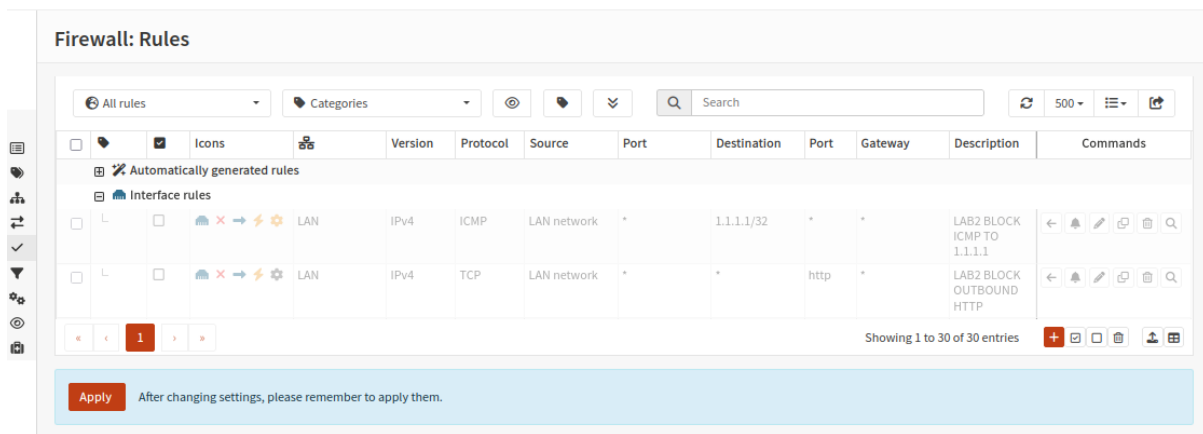
Search

☐			Version	Protocol	Source	Port	Destination	Port	Translate S...	Transl...
Automatically generated rules										

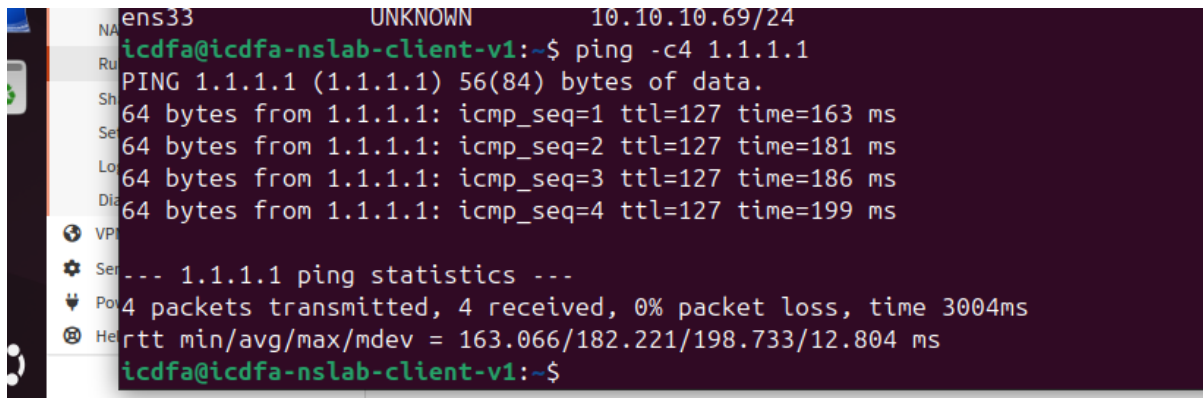
Apply

Brief Explanation: This shows that the OPNsense is configured to use Automatic Source NAT rule generation

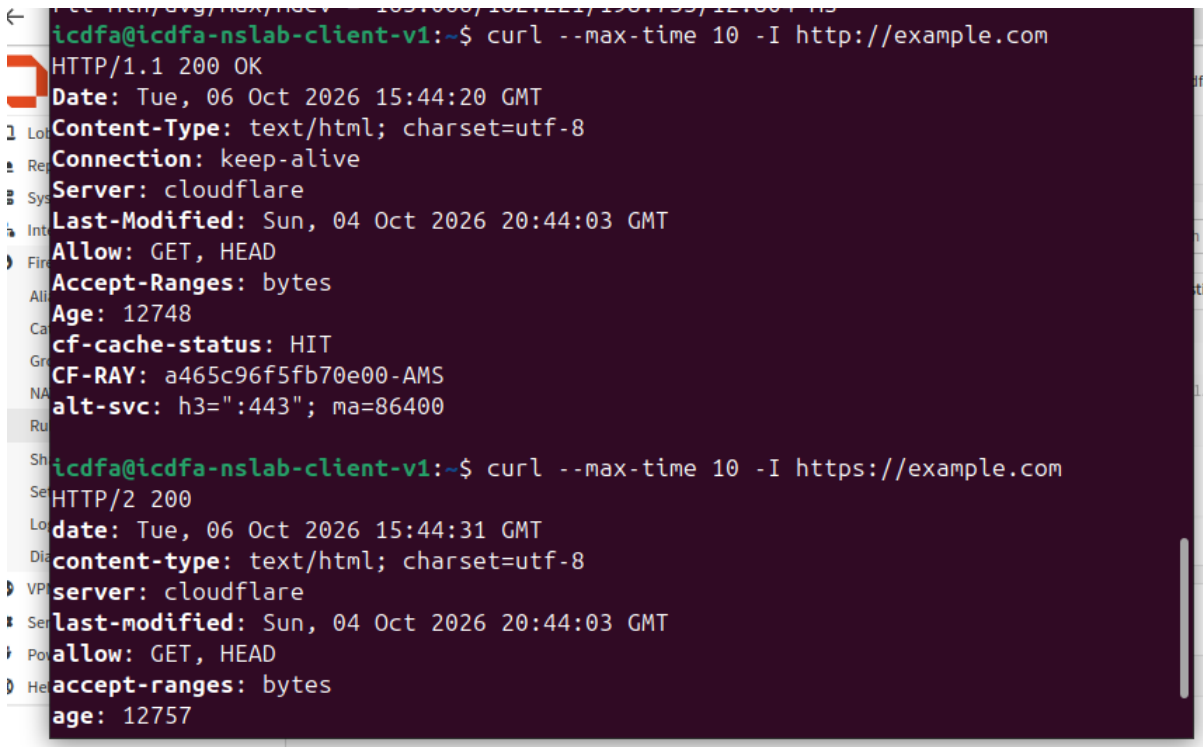
PART H: RESTORING THE LABIRATORY



Brief Explanation: This shows that the two block rule were disable and we expect the former command to be able to run successfully



Brief Explanation: This shows that now client can ping the internet after disabling the rule



Brief Explanation: This shows that now client can request on HTTP website

QUESTIONS

1. Why must the specific block rules be placed above the broad allow rule?

Ans: because when the firewall wanted to filter the traffic if the allow is at above the block will not be execute because allow all will override it, therefore it has to be below so that when the specific policy is blocked, the allow rule will now allow the remaining rules

2. Which five packet attributes are most useful when explaining a firewall decision?

Ans:

Protocol: for example, TCP, UDP, or ICMP

Source IP address: where the packet came from

Destination IP address: where the packet is going

Source port: the originating application

Destination port: the service being requested, such as TCP 80 for HTTP or 443 for HTTPS

3. Why did blocking ICMP not block HTTPS?

Ans: Because HTTPS uses TCP protocol therefore blocking ICMP cannot affect HTTPS

4. What difference did you observe between the blocked TCP port 80 traffic and permitted TCP port 443 traffic?

Ans: the difference is that we specify a destination HTTP which uses port 80 there fore our rule will only affect port 80 without affecting port 443 unless we specified it to destination Any

5. What role does outbound NAT play when icdfa-nslab-client-v1 uses a private IPv4 address?

Ans: the outbound NAT translate the source port no. of the client from 56380 to 9063 in other for it to communicate with the internet

6. Why is restoring the original state an important part of a controlled security laboratory?

Ans: Because it removes temporary configuration changes made during the experiment